ML for Computer vision: models, learning and inference

Adapted from Simon Prince’s book

http://www.computervisionmodels.com/

ML for Computer Vision

- Computer vision models
  - Two types of model
  - Regression
  - Classification
- Which type should we choose?
- Applications

Computer vision models

- Observe measured data, **x**
- Draw inferences from it about state of world, **w**

Examples:

- Observe adjacent frames in video sequence
- Infer camera motion

- Observe image of face
- Infer identity

- Observe images from two displaced cameras
- Infer 3d structure of scene

Regression vs. Classification

- Observe measured data, **x**
- Draw inferences from it about world, **w**

When the world state **w** is continuous we'll call this regression

When the world state **w** is discrete we call this classification

Ambiguity of visual world

- Unfortunately visual measurements may be compatible with more than one world state $\mathbf{w}$
  – Measurement process is noisy
  – Inherent ambiguity in visual data
- Conclusion: the best we can do is compute a probability distribution $Pr(\mathbf{w}|\mathbf{x})$ over possible states of world

Refined goal of computer vision

- Take observations **x**
- Return probability distribution $Pr(\mathbf{w}|\mathbf{x})$ over possible worlds compatible with data

(not always tractable – might have to settle for an approximation to this distribution, samples from it, or the best (MAP) solution for **w**)

Components of solution

We need

- A model that mathematically relates the visual data $\mathbf{x}$ to the world state $\mathbf{w}$. Model specifies family of relationships, particular relationship depends on parameters θ

- A learning algorithm: fits parameters θ from paired training examples $\mathbf{x}_i,\mathbf{w}_i$

- An inference algorithm: uses model to return $Pr(\mathbf{w}|\mathbf{x})$ given new observed data $\mathbf{x}$.

Types of Model

The model mathematically relates the visual data $\mathbf{x}$ to the world state $\mathbf{w}$. Two main categories of model

1. Model contingency of the world on the data $Pr(\mathbf{w}|\mathbf{x})$
2. Model contingency of data on world $Pr(\mathbf{x}|\mathbf{w})$

Generative vs. Discriminative

1. Model contingency of the world on the data $Pr(\mathbf{w}|\mathbf{x})$

   (DISCRIMINATIVE MODEL)

2. Model contingency of data on world $Pr(\mathbf{x}|\mathbf{w})$

   (GENERATIVE MODELS)

Generative as probability model over data and so when we draw samples from model, we GENERATE new data

Type 1: Model $Pr(\mathbf{w}|\mathbf{x})$ - Discriminative

How to model $Pr(\mathbf{w}|\mathbf{x})$?

1. Choose an appropriate form for $Pr(\mathbf{w})$
2. Make parameters a function of $\mathbf{x}$
3. Function takes parameters θ that define its shape

Learning algorithm: learn parameters θ from training data $\mathbf{x},\mathbf{w}$

Inference algorithm: just evaluate $Pr(\mathbf{w}|\mathbf{x})$

Type 2: Pr(**x**|**w**) - Generative

How to model Pr(**x**|**w**)?

1. Choose an appropriate form for Pr(**x**)
2. Make parameters a function of **w**
3. Function takes parameters θ that define its shape

Learning algorithm: learn parameters θ from training data **x**,**w**

Inference algorithm: Define prior Pr(**w**) and then compute Pr(**w**|**x**) using Bayes' rule

$$Pr(\mathbf{w}|\mathbf{x}) = \frac{Pr(\mathbf{x}|\mathbf{w})Pr(\mathbf{w})}{\int Pr(\mathbf{x}|\mathbf{w})Pr(\mathbf{w})d\mathbf{w}}$$

Summary

Two different types of model depend on the quantity of interest:

1. **Pr(w|x)** Discriminative
2. **Pr(x|w)** Generative

Inference in discriminative models easy as we directly model posterior Pr(**w**|**x**). Generative models require more complex inference process using Bayes' rule

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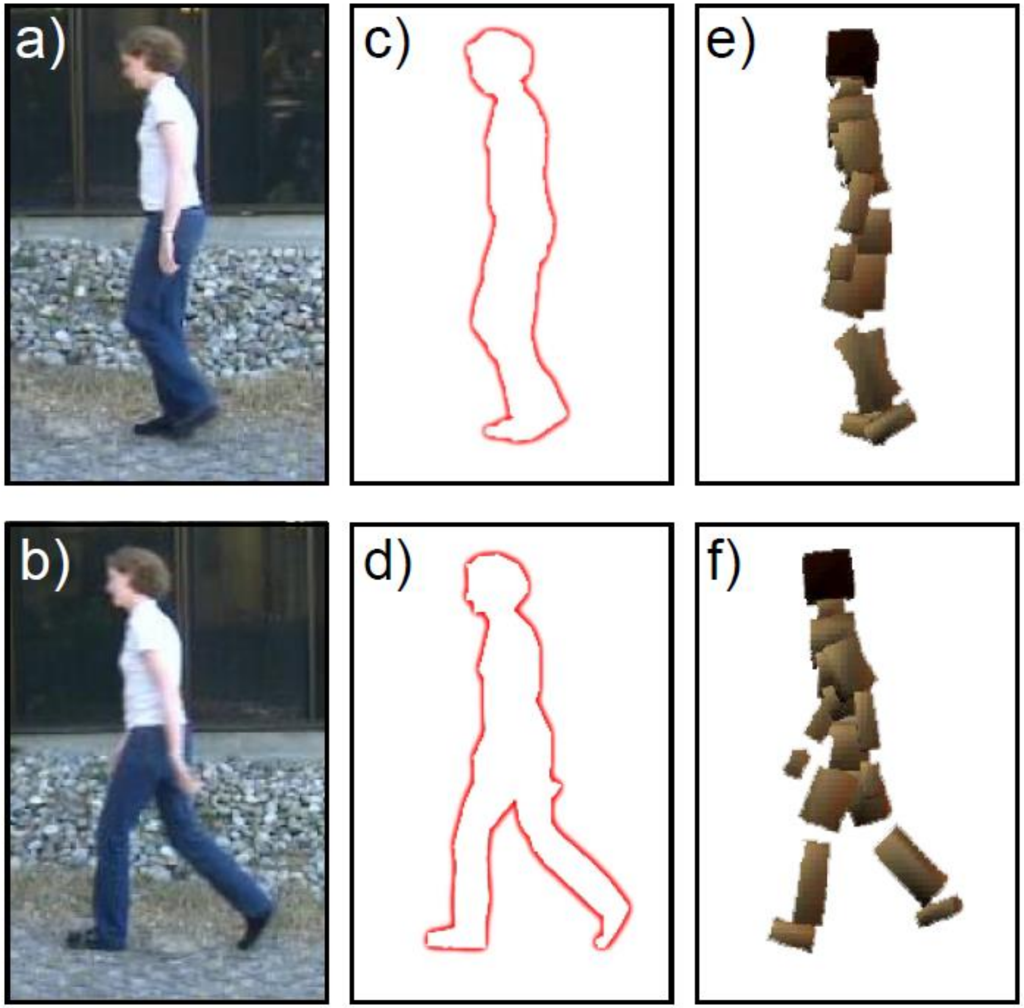
Regression

Consider simple case where

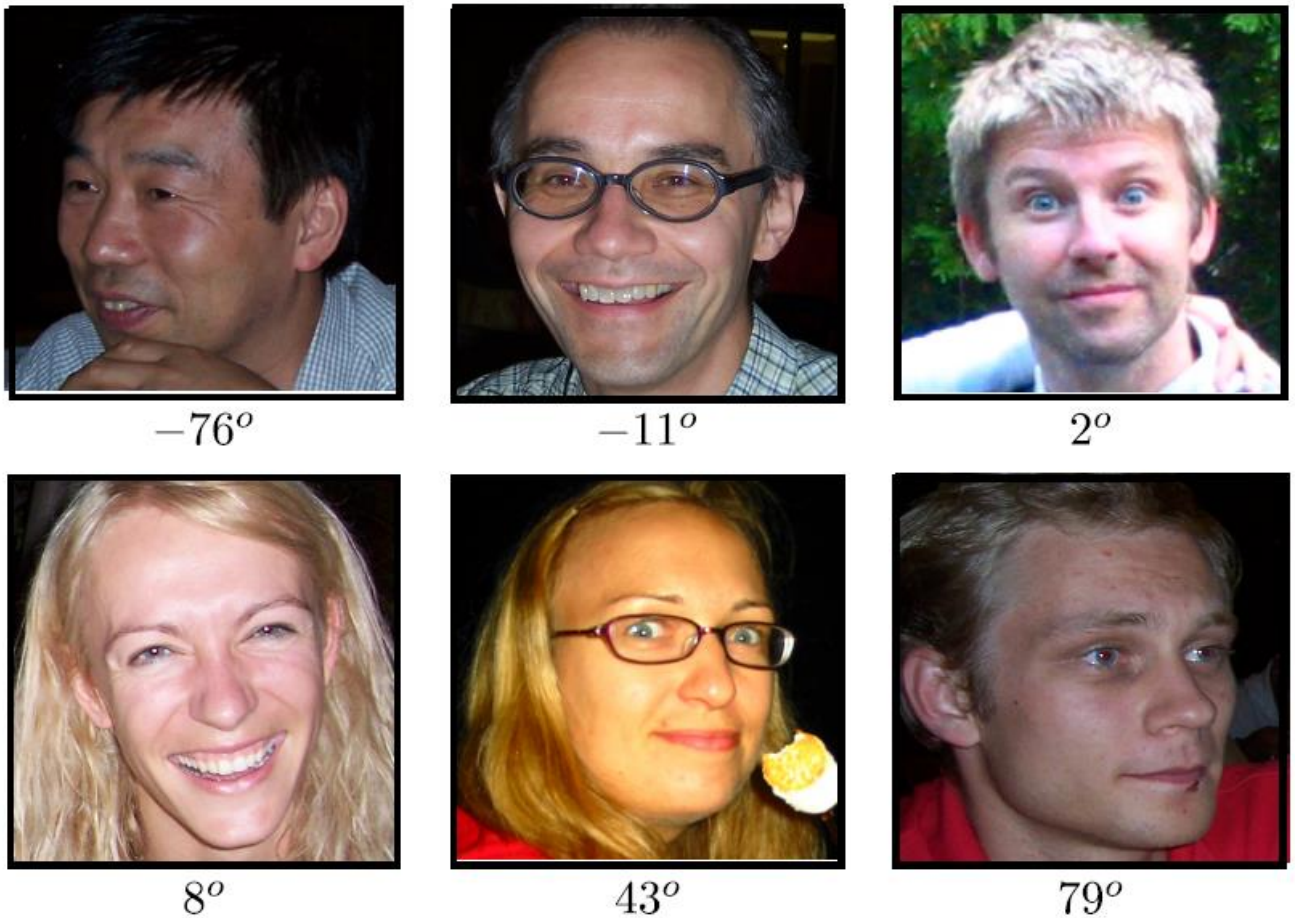
- we make a univariate continuous measurement **x**
- use this to predict a univariate continuous state **w**

(regression as world state is continuous)

Regression application 1: Pose from Silhouette



Regression application 2: Head pose estimation



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Classification

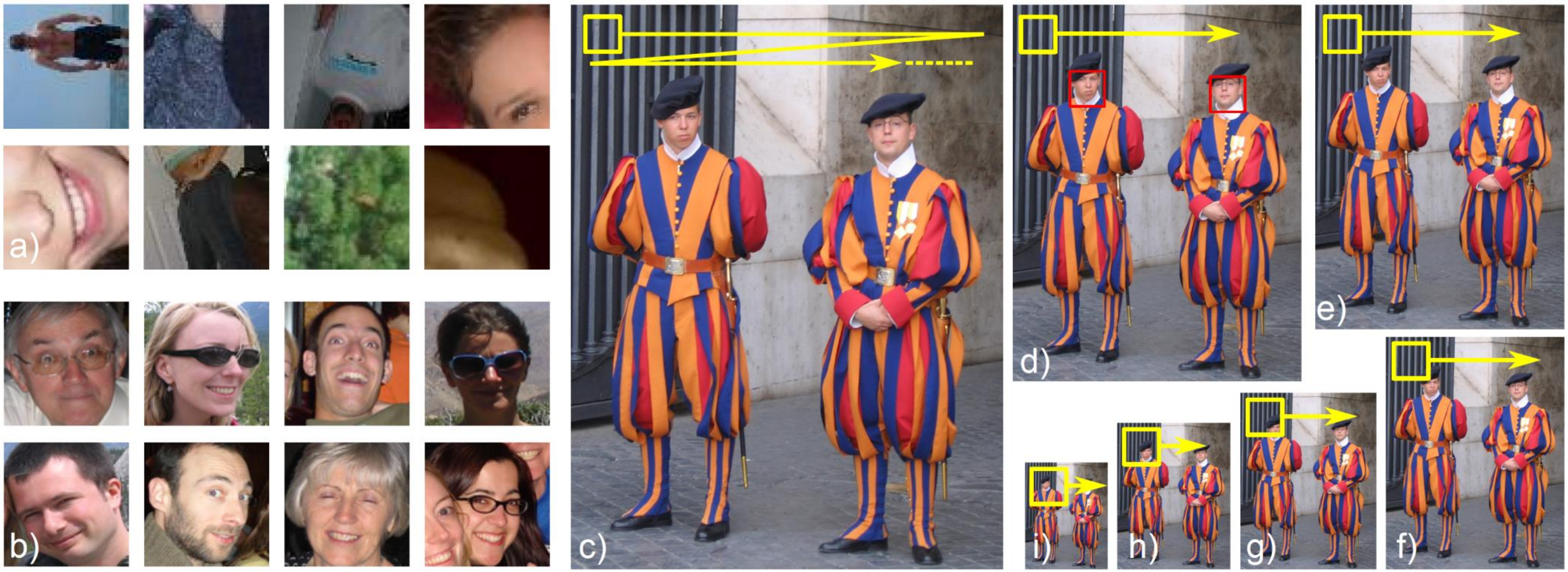
Consider simple case where

- we make a univariate continuous measurement x
- use this to predict a discrete binary world

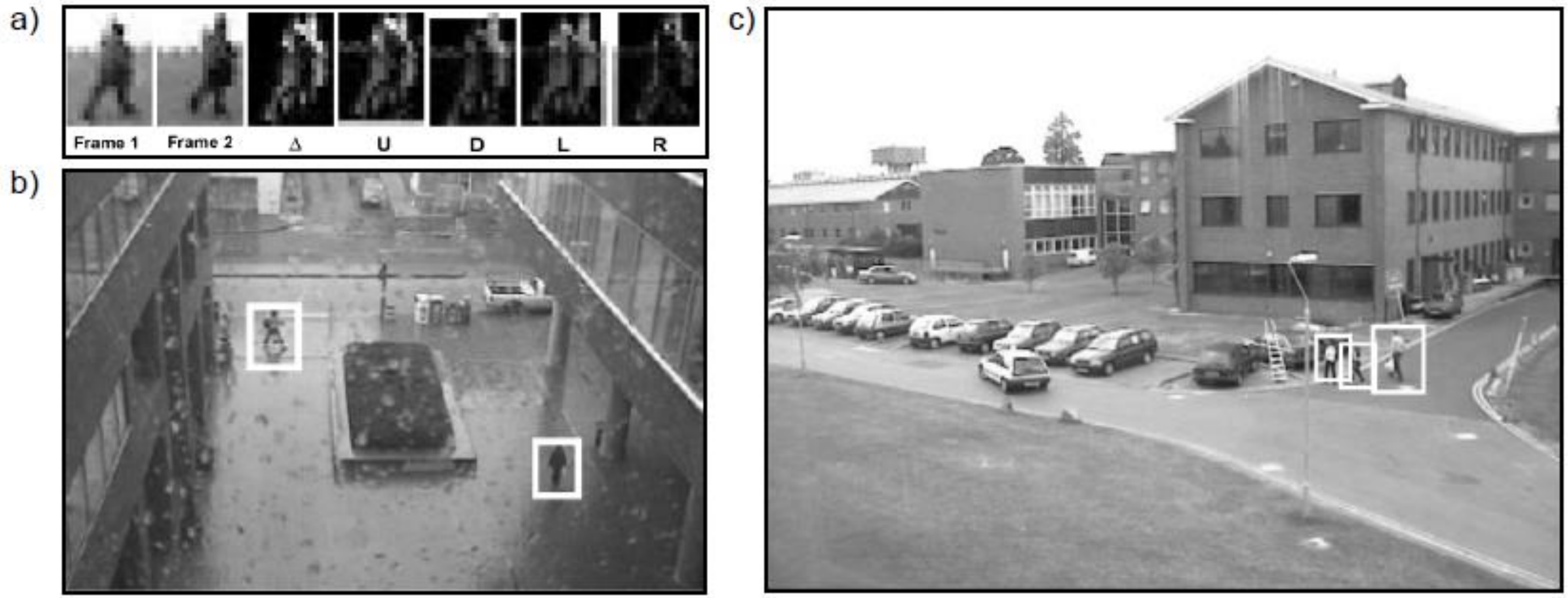
$$w \in \{0, 1\}$$

(classification as world state is discrete)

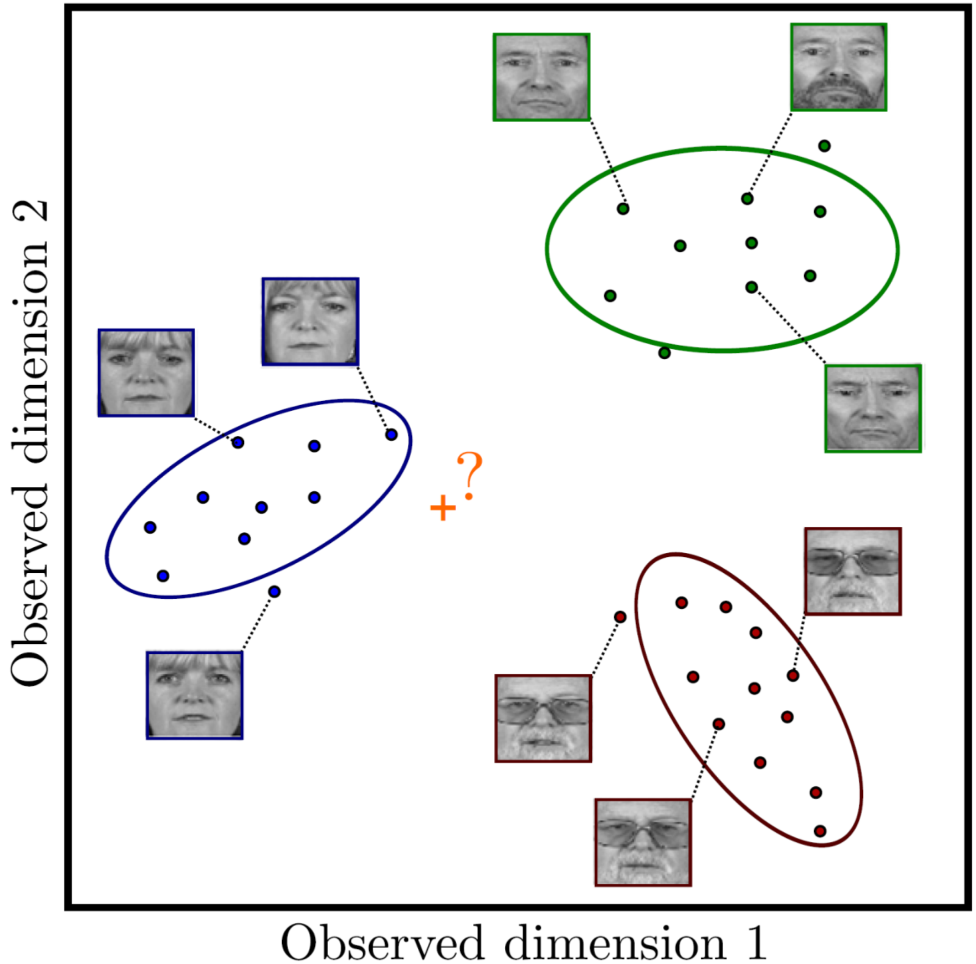
Classification Example 1: Face Detection



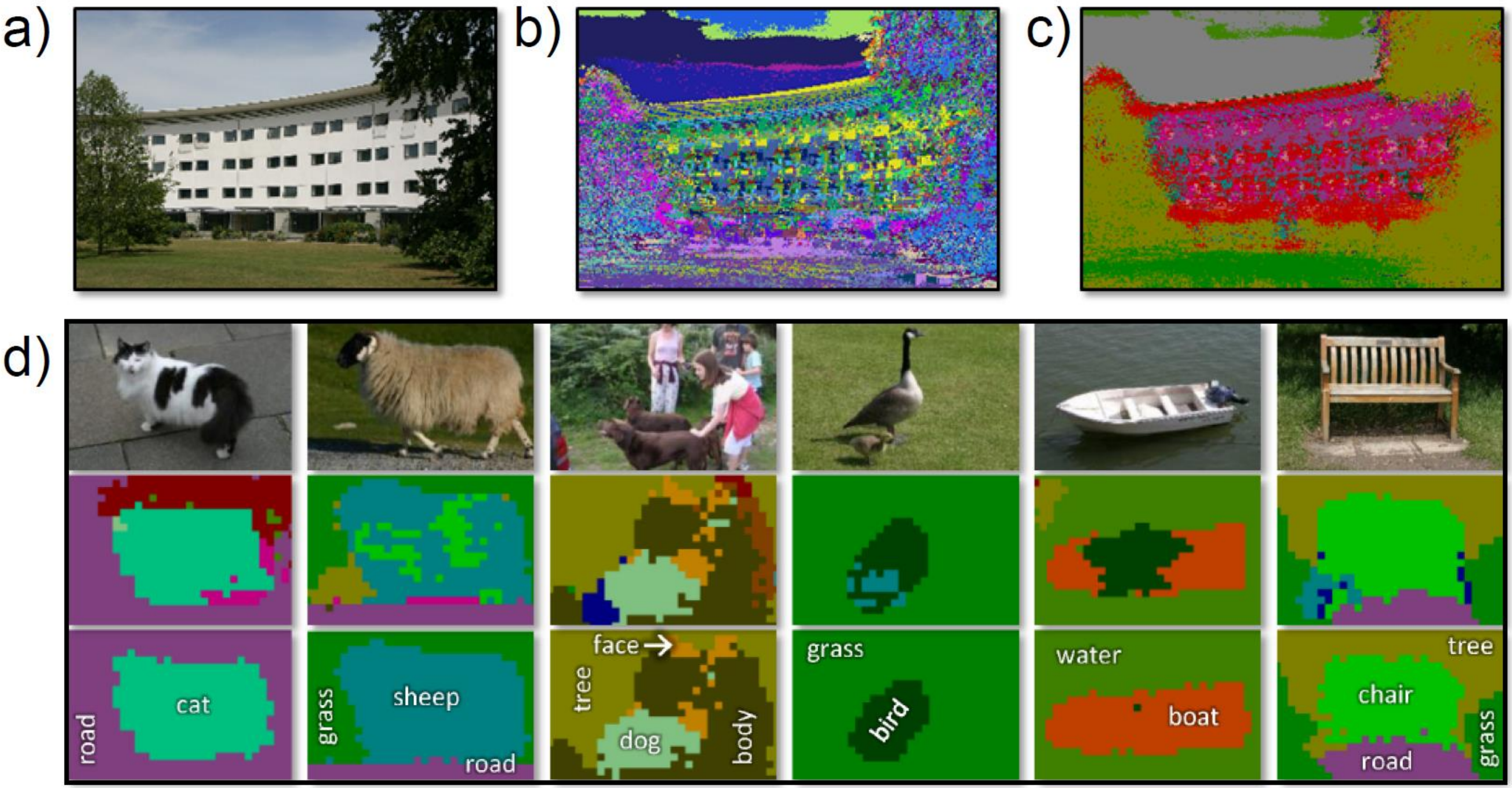
Classification Example 2: Pedestrian Detection



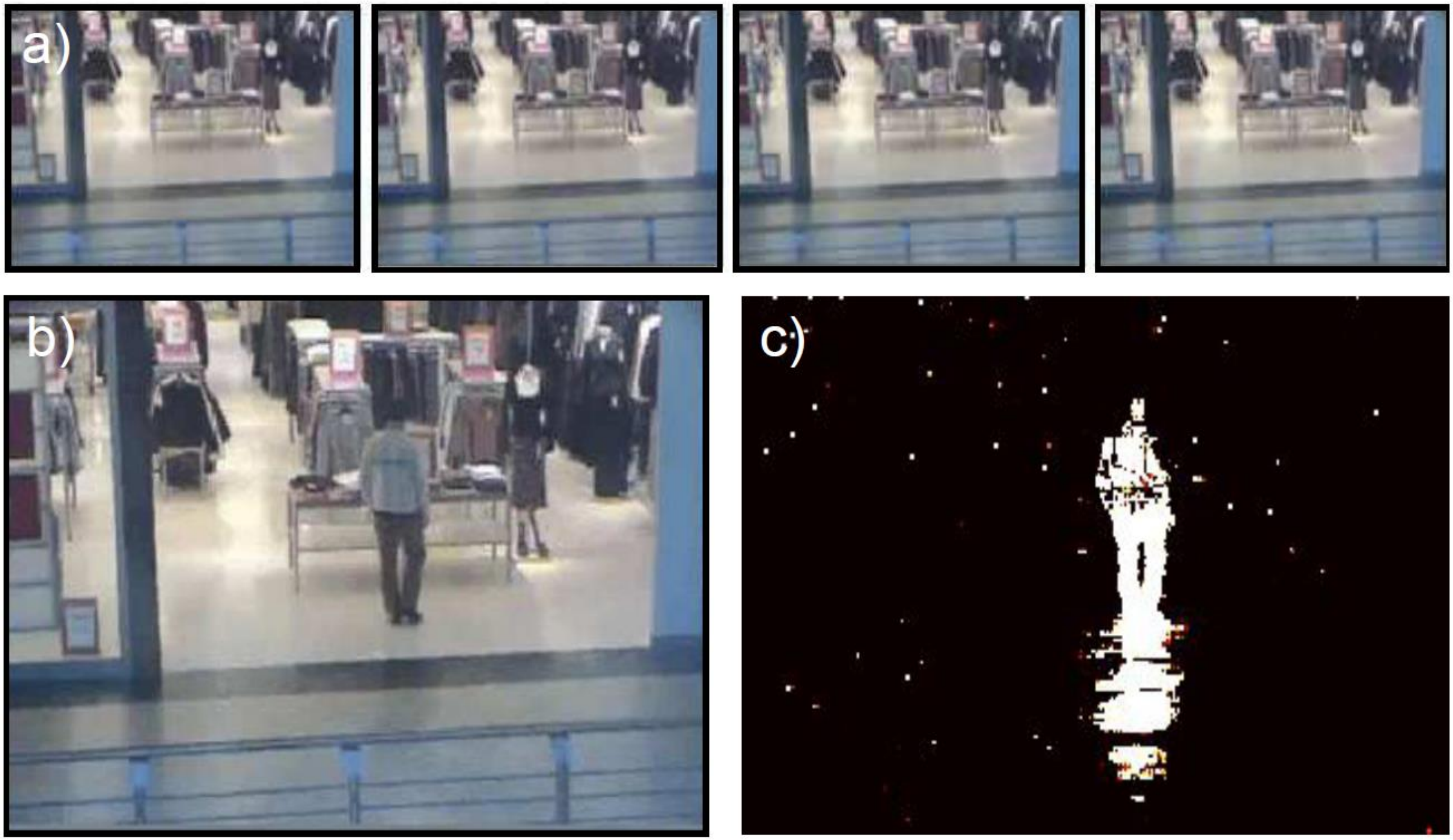
Classification Example 3: Face Recognition



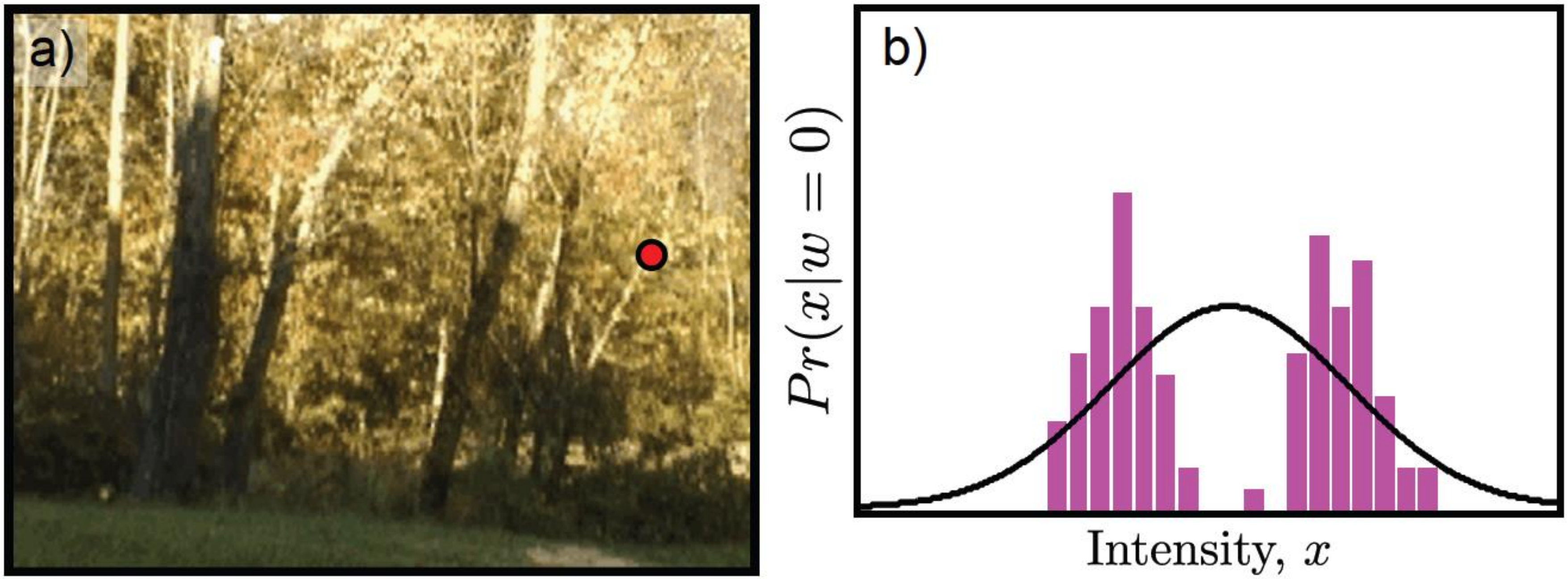
Classification Example 4: Semantic Segmentation



Application: Background subtraction



Application: Background subtraction



But consider this scene in which the foliage is blowing in the wind. A normal distribution is not good enough! Need a way to make more complex distributions

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Which type of model to use?

1. Generative methods model data – costly and many aspects of data may have no influence on world state
2. Inference simple in discriminative models
3. Data really is generated from world – generative matches this
4. If missing data, then generative preferred
5. Generative allows imposition of prior knowledge specified by user

Structure

- Computer vision models
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Summary

- Seen two types of model

	Model $Pr(w\|x)$	Model $Pr(x\|w)$
Regression $x \in [-\infty, \infty], w \in [-\infty, \infty]$	Linear regression	Linear regression
Classification $x \in [-\infty, \infty], w \in \{0, 1\}$	Logistic regression	Probability density function

  - Probability density function
  - Linear regression
  - Logistic regression